

Homework 1

CS 09410 and CS 09510

Topics: Chapter 1

Due February 8, 11.59 pm

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Read the following instructions before submitting your work:

- Submitting work copied (in part or in whole) from other persons, websites, or generative AI tools such as ChatGPT is strictly forbidden.
- Submit your answers on Canvas as a single PDF file.
- Include a heading with your name and assignment number/date.
- Handwriting is allowed if writing is readable by others.

Remarks: 1 byte is 8 bits. bps stands for bits per second; Bps, with uppercase B, stands for bytes per second. Letters k, M, and G denote powers of 10: k is 10^3 , M is 10^6 , and G is 10^9 . That is, 1 kbps = 8 kbps = 8,000 bps.

Problem 1: Describe the most popular wireless *access* technologies today. Compare and contrast them.

There are *very* two popular wireless Internet access technologies today:

- a. Wifi (802.11) In a wireless LAN, wireless users transmit/receive packets to/from a base station (i.e., wireless access point) within a radius of few tens of meters. The base station is typically connected to the wired Internet and thus serves to connect wireless users to the wired network.
- b. 3G, 4G, and 5G wide-area wireless access networks. In these systems, packets are transmitted over the same wireless infrastructure used for cellular telephony, with the base station thus being managed by a telecommunications provider. This provides wireless access to users within a radius of tens of kilometers of the base station.

Problem 2: Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are R_1 and R_2 , respectively. If the switch uses store-and-forward packet switching, what is the total end-to-end *transmission* delay for a packet of length L bits?

Topology: source host — R_1 — switch — R_2 — destination host

1st hop: delay = $\frac{L}{R_1}$

2nd hop: delay = $\frac{L}{R_2}$

End-to-end delay = $\frac{L}{R_1} + \frac{L}{R_2}$ seconds

Problem 3: Suppose users share a 2 Mbps link; each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20% of the time.

- a) When circuit switching is used, how many users can be supported? 2 users can be supported because each user requires half of the link bandwidth.
- b) Assume packet switching is used. Will there be any significant queuing delay if two or fewer users transmit simultaneously? There won't be significant queuing delay if two or fewer users transmit simultaneously. What if three users transmit simultaneously? Yes. The incoming traffic would be higher than the outgoing capacity, resulting in queuing delay. (You can easily show that traffic intensity ≥ 1).
- c) Suppose now there are three users. Find the probability that at any given time, one user is transmitting. $p = 0.2$. The probability of having one user transmitting at any given time is given by $\binom{3}{1} \times p^1 \times (1 - p)^{3-1} \approx 0.384$.
- d) Now find the probability that all three users are transmitting simultaneously. $p = 0.2$. $\binom{3}{3} \times p^3 \times (1 - p)^{3-3} = 0.2^3 = 0.008$ (or 0.8%)

Problem 4: Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays tend to be constant and which tend to be variable?

The delay components are processing delays, transmission delays, propagation delays, and queuing delays. All of these delays are fixed, except for the queuing delays, which are variable.

Problem 5: How long does it take to transmit a packet of length 1,000 bytes over a link of distance 2,500 km, propagation speed 2.5×10^8 m/sec, and transmission rate 2 Mbps?

Assuming that queuing and processing delays are negligible,

$$t_{\text{prop}} = \frac{2.5 \times 10^6 \text{ m}}{2.5 \times 10^8 \text{ m/sec}} = 10 \text{ ms}$$

$$t_{\text{trans}} = \frac{8,000 \text{ bits}}{2 \times 10^6 \text{ bits/sec}} = 4 \text{ ms.}$$

Total delay = 14 ms.

Problem 6: Suppose Host A wants to send a large file to Host B. The path from A to B has three links of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps.

- a) Assuming no other traffic in the network, what is the throughput for the file transfer?
500 kbps.
- b) What if R_2 is reduced to 100 kbps? 100 kbps.

Problem 7: What are the five layers in the Internet protocol stack? Briefly describe the main responsibilities of each of these layers. The five layers in the Internet protocol stack are – from top to bottom – the application layer, the transport layer, the network layer, the link layer, and the physical layer. The principal responsibilities are outlined in Section 1.5.1.